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NL East	NL Central	NL West
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20262025

AL

BAL CHW ATH
 BOS CLE HOU
 NYY DET LAA
 TBR KCR SEA
 TOR MIN TEX

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ATL CHC ARI
 MIA CIN COL
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Positional Adjustment

by [Piper Slowinski](#)

February 26, 2010

In essence, the positional adjustment is a correction to account for the fact that different positions are more challenging than others, which is a pretty easy thing to accept. An average center fielder is worse than an average first baseman. A position might be more challenging because there's a higher volume of plays or a higher degree of difficulty.

We need an adjustment because we currently measure defense using [Ultimate Zone Rating \(UZR\)](#) or [Defensive Runs Saved \(DRS\)](#) which are created by comparing a player to average performance at his position. Those numbers show up as runs above or below average at a given position. So a shortstop who is five runs better than the average shortstop gets a +5 and a left fielder who is five runs better than the average left fielder gets a +5.

We compare players to their positional averages because we want to base player evaluations on a similar set of plays. It would be silly to compare the conversion rate of a third baseman against that of a right fielder because not only do they field a different number of balls, they're just a totally different kind of batted ball. As a result, when you look at the initial defensive statistic, you're looking at only how that player compares to their fellow position-mates.

And when it comes to overall player evaluation and WAR, we need some way to balance out those differences in position. We need to find a way to account for the difference between the positions. The positional adjustment does just that.

In general, we want to add runs for players who play tough positions and subtract runs for players who play easier positions to account for the fact that average at one does not equal average at the other in terms of total run prevention. For example, for a shortstop, the adjustment is +7.5 runs per full season. For a left fielder, it's -7.5 runs per full season.

That means the difference between an average left fielder and an average shortstop is about 15 runs per season. That's a sizable gap. The positional adjustment is a method for putting a bunch of different positions onto one uniform scale so that when you add the adjustment to a player's runs saved to get what we call "DEF," meaning Defensive Runs Above Average, you can compare players no matter their position.

The actual numbers we use are based on [some calculations that were done about a decade ago](#) that used the performance of players who moved positions. It's certainly reasonable to suggest that those numbers have changed as the game has changed, so use the adjustments as guides more than as firm rules. The DH adjustment might be too negative because it's harder to hit when you're not playing in the field, and the catcher adjustment might be a bit too large. There's lots of room to disagree on the precise decimals and if you're so inclined, I'd invite you to come up with a more accurate rendering of the numbers. [Here is one recent example of such work.](#)

One of the biggest challenges in this effort is that players aren't assigned positions at random and don't switch positions at random, meaning we have to get around the likely bias. For example, left-handed players don't play shortstop, so the very best left-handed defenders wind up at other spots even if a right-hander that talented would play short. We all recognize that these are estimates and you could certainly slide them around a little if you think we're wrong.

But the important thing is that we need a positional adjustment of some kind. We calculate defense based on position average but the position averages need to be corrected so that an average catcher and an average first baseman don't look the same. The positional adjustment does this for us. It might make sense to come up with a better version of the numbers we use, but adjusting for position is a vital piece of overall player analysis.

Positional adjustments are calculated based on a full 162 games, which equates to 1,458 defensive innings. So if a first baseman plays 1,214 innings with -12.5 positional adjustment for a full season, his adjustment for that period will be -10.4 runs. For players who play multiple positions, it's simply the adjustment at each position added together.

FanGraphs Positional Adjustment	
Position	Full Season Adjustment
C	+12.5
1B	-12.5
2B	+2.5
SS	+7.5
3B	+2.5
LF	-7.5
CF	+2.5
RF	-7.5
DH	-17.5
Per 1,458 innings (162 defensive games)	

Further Reading

[The Beginner's Guide to the Positional Adjustment – FanGraphs](#)

[Replacement Level](#)

[Splits](#)

Piper was the editor-in-chief of [DRaysBay](#) and the keeper of the [FanGraphs Library](#).

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4 COMMENTS

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Six ⌚ 6 years ago

Tango Tiger and others made a huge mistake in formulating Fielding WAR adjustments based on position... and to my knowledge nobody has called it out.

The current state is wrong on so many counts it is hard to believe so-called stat professionals could get it so screwed up.

It will take an essay to understand why (properly).

I'll post it soon enough.

Get ready for a real shock.

👍 -8 🗨️

CharlieC ⌚ 6 years ago

| 🗨️ *Reply to* [Six](#)

Positional adjustment is not a fielding adjustment.

👍 5 🗨️

larrooo **Member since 2018** ⌚ 5 years ago

Asking for a friend ...

Why are all positions created equal?

Why are you adjusting to an 'average' player? Shouldn't we adjust to the hypothetical replacement player?

If the defense of the best fielding shortstop helps his team win more than the defense of the best fielding first baseman ... then does the defense of the worst fielding shortstop hurt his team more than the defense of the

worst field first baseman? Why award that worst fielding shortstop with the same positional adjustment as the best fielding shortstop?

👍 2 🗨️

phoenixone ⌚ 5 years ago

| 🗨️ Reply to [larrooo](#)

It seems as though you misunderstand what goes into WaR. Shortstop is a more defensively taxing position than, say, First Baseman. OAA is meant to measure defensive quality, but it is significantly easier to get +5 OAA at First Base than at Shortstop, and this is an attempt to correct for that.

👍 -2 🗨️

Updated: Friday, October 2, 2026 2:17 AM ETUpdated: 10/2/2026 2:17 AM ET

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Mitchel Lichtman

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